

The challenge is to develop and deploy creative applications, on-demand, over our private Cloud infrastructure. These applications should be accessible from anywhere and via any web browser, without the need for users to install, maintain, or troubleshoot them. They will be hosted and managed entirely by our solution, ensuring smooth functionality, even with high traffic, through built-in auto-scaling capabilities.

In addition to these objectives, the application must integrate Artificial Intelligence (AI) models to either enhance client services or improve the application's overall performance. The AI models can be applied in various ways, including:

1. Client-Centric AI Features:

- a. **Recommendation Systems:** Use machine learning algorithms to suggest personalized services or content to clients based on their historical interactions and preferences.
- b. **Predictive Analytics:** AI models can predict client needs or behaviors, allowing the application to pre-emptively offer services or support based on likely future actions.
- c. **Chatbots and Virtual Assistants:** AI-powered chatbots can assist clients by understanding natural language queries and providing real-time support or guidance.

2. AI for Application Enhancement:

- a. **Performance Optimization:** AI models can monitor and analyze application performance metrics, automatically adjusting resources for optimal performance, especially during peak usage.
- b. **Security and Anomaly Detection:** Implement AI-based security models that detect unusual patterns or behaviors in real time to prevent data breaches or system vulnerabilities.
- c. **Automated Scaling and Resource Management:** Use AI to predict traffic loads and automatically scale resources, ensuring smooth operations even under heavy demand.

Each client, organization, or group of users will have access to their own instance of the application, ensuring isolation and privacy. To support this, the infrastructure will include solutions that make software deployment easy, rapid, and compatible across any hardware or operating system.

Clients require AI models to be integrated into the application in a way that best suits their needs, ensuring that AI enhances their overall experience while also improving the application's efficiency, security, and scalability.



B. Project Objectives

The first goal of the project is to provide virtualized computing resources on demand. These resources can be servers, storage spaces or containers. The infrastructure solution should also be able to adapt to workload changes by provisioning and de-provisioning resources automatically.

Additionally, the chosen infrastructure solution should:

- Be an open-source solution.
- Have a modular/ multi-tenant architecture.
- Provide orchestration capabilities.
- Support a large choice of hypervisors.
- Support private and hybrid deployment.
- Leverages commodity hardware.
- Provide self-healing, reliable, data redundancy protection from failures.
- Scale vertically and horizontally distributed storage.
- Backup and archive large amounts of data with linear performance.
- Easily add capacity.
- Support optimized and efficient resource usage.
- Provide a balanced and intelligent distribution of resources.
- Deliver simple management of cloud resources through a centralized dashboard.
- Provide standards for building open, massively scalable clouds.
- Provides native Load Balancer.
- Integrate floating IP assignment.

To be able to deliver virtualized resources, a virtualization solution that respects these criteria must be installed:

- Fully supported and compatible with the infrastructure management solution
- Can use a wide variety of certified Linux-supported hardware platforms.
- Supports live migration
- Scales to match demand load if the number of guest machines and requests increases
- Allows the most demanding application workloads to be virtualized

Also, the project should propose a solution to minimize infrastructure resources provisioning time, eliminate repetitive tasks and reduce mistakes & errors risk. The proposed solution will be used for managing the entire lifecycle of infrastructure and applications within the private cloud (to deploy needed services and platforms over the computing infrastructure). It should also be:

- Open-source software.
- Facilitates a smooth connection with the foundational infrastructure.
- Uses templates that describe the infrastructure for a cloud application in a text file that is readable and writable by humans and can be checked into version control.
- Can be used to specify the relationships between resources (e.g. this volume is connected to this server).



- Should allow integration with other components of the cloud infrastructure. It will allow creation of most resource types (such as instances, floating ips, volumes, security groups, users, etc.), as well as some more advanced functionality such as instance high availability, instance auto scaling, and nested stacks.

In addition, the project should include solutions to provide the deployment environment required to run the web application. Since each client will have his own instance of the application, resource optimization, greater efficiency and reducing overhead should be taken into consideration. Portability also should be ensured since it will be possible to deploy the application easily to multiple different operating systems and hardware platforms.

Centralized management and orchestration for all the deployment environments must be provided too. It will simplify the management of all the application instances and, simultaneously, make it more efficient. This orchestration solution should:

- Installable almost on any Linux distribution such as Debian or Ubuntu.
- Installable automatically using configuration management tools.
- Be standard and be available on many platforms.
- Facilitate both declarative configuration and automation.
- Full integration with the cloud infrastructure services, specifically in terms of identity management, networking and block storage.
- Able to deliver deployment environments as a service.
- Based on standard solutions and available on many platforms.
- Facilitate both declarative configuration and automation.
- Provide deployment patterns and templates.
- Provide service discovery and load balancing.
- Ensure self-healing.

To deploy the orchestration solution, we must use a solution that meets the following criteria:

- Open-source software.
- Able to provide provisioning, configuration management, and application-deployment features.
- To run on any Windows, Linux and Unix-like systems.
- Agentless, does not require to deploy any agents to nodes.
- Using simple and standardized communication protocols like SSH.

Moreover, you must focus on one of two key aspects of managing and operating a cloud-based infrastructure:

- **Data Backup and Recovery:** To ensure the integrity and availability of essential configuration data for a stable orchestrated environment, this task involves implementing a robust backup and recovery mechanism for the core data store responsible for managing system state and configurations. This process will include setting up regular data backups



and establishing a reliable method to restore data in the event of failure or corruption.

- **In-Place Environment Version Upgrades:** Maintaining an up-to-date orchestrated environment without service interruptions is essential for long-term reliability. This task involves focusing on upgrading the environment version while ensuring continuous availability. The objective is to plan and implement an upgrade strategy that minimizes downtime and preserves compatibility across all components.

These two scenarios are crucial for certain clients, and it is essential to prepare and test at least one of them.

Finally, to ensure effective project management, it should propose a solution for effectively observing and visualizing the system's performance and health. This is essential, as observation and visualization serve as critical tools for identifying and diagnosing potential issues that could impact the system's reliability, in cases of anomalies or problems. It's imperative to automate these key tasks. The chosen configuration management tool should prioritize using simple and standardized communication protocols, such as SSH, to ensure ease of interaction and minimize complexity. It's also worth noting that the tool should be agentless, meaning it doesn't require the installation of additional components on target systems. These qualities simplify the automation process further, contributing to the efficiency and effectiveness of the project's management and maintenance.

The choice of the configuration management tool is crucial for this task, and it must meet several essential requirements extracted from the keywords in the project brief.

- **Real-Time Data Collection:** The configuration management tool should enable the configuration and deployment of components capable of collecting real-time data from the entire infrastructure, including servers, storage, and containers.
- **Dashboard Generation:** It must be capable of creating and maintaining configurations of customized dashboards to visualize essential metrics and statistics for infrastructure management.
- **Alerts and Notifications:** The tool should provide capabilities for configuring custom alerts and real-time notifications in the event of threshold breaches or critical issues.
- **Scalability:** It should allow for the automatic addition or removal of monitoring resources to adapt to variations in workload.
- **Security:** The tool must ensure an appropriate level of security by maintaining data confidentiality and providing robust authentication and authorization mechanisms.



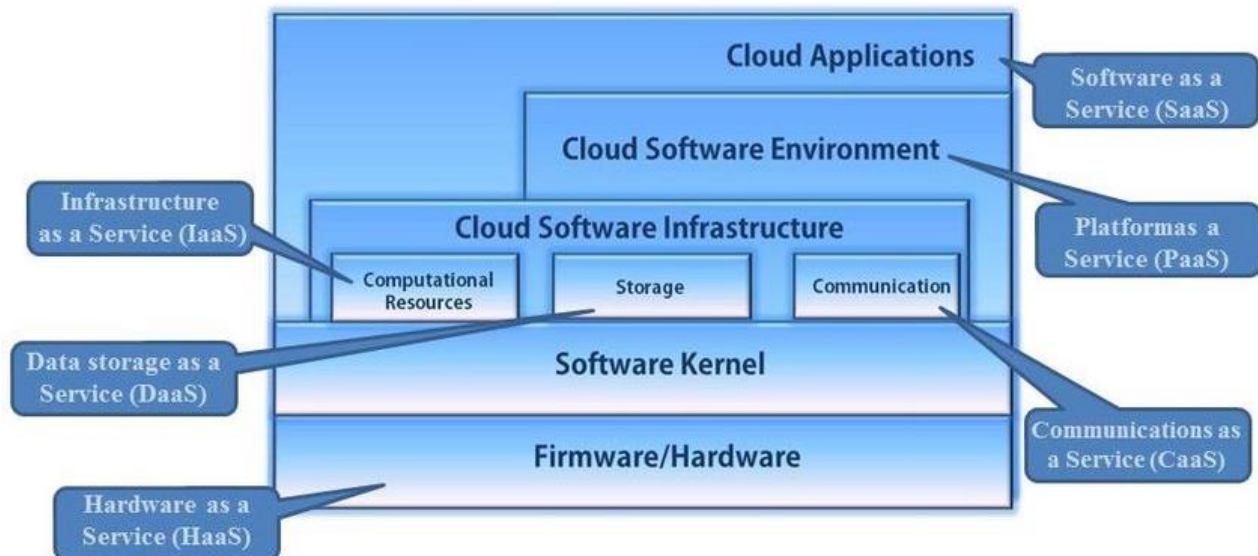
C. Software Basic Functional Features Description

The software should be a web-based application highly available. that will be deployed over the computing infrastructure.

D. Architecture Overview

On an indicative basis, in addition to the web software development, the proposed solution should offer:

- The infrastructure layer that will deliver computing and storage resources such as servers, containers or disks.
- The management, orchestration and automation services are required to optimize resources utilization, reduce provisioning time and improve system reliability.



E. Schedule & Deadline

The global project duration is about 20 weeks. During this period, several deliverables must be submitted in accordance with this planning:

- **Week 3:** Global infrastructure design and detailed project schedule.
- **Week 8:** Test of the computing infrastructure.
- **Week 12:** Test of the software deployment environment.
- **Week 19:** Present the developed application.
- **Week 21:** The Deployment of the application.

F. Additional Information

- A detailed specification of the functionalities that the software must provide should be provided by the end of phase 3. The global application idea will be discussed all along the first 3 phases with the project supervisors.
- All documents related to this project must be written in English.
- Clients have the right to add additional services and change chosen technical choices.

